

# OnlineCA.jl: A Julia Package for Out-of-core and Sparse Correspondence Analysis

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## Summary

Correspondence Analysis (CA) is a multivariate technique for the exploratory analysis of contingency tables (Greenacre, 2017; Hill, 1974). Like Principal Component Analysis (PCA), CA decomposes a data matrix into low-dimensional row and column scores, but it operates on a standardized residual matrix that compares observed counts with the expectation under independence. CA has been widely applied across diverse fields including ecology (Braak, 1986), linguistics and text analysis (Greenacre, 2017), market research (Greenacre, 2017), and more recently single-cell omics (Hsu & Culhane, 2023).

Despite its broad applicability, CA becomes computationally prohibitive for large data matrices, since it requires the singular value decomposition (SVD) of an  $N \times M$  matrix. In particular, recent advances in single-cell omics have led to datasets with millions of cells, for which standard CA implementations often fail to scale. To meet this requirement, I present `OnlineCA.jl`, a Julia package for out-of-core and sparse CA (<https://github.com/chiba-ai-med/OnlineCA.jl>). The same out-of-core machinery is reused to support Multiple Correspondence Analysis (MCA) on large categorical tables, including the standard Benzécri (Benzécri, 1979) and Greenacre (Greenacre, 2017) eigenvalue corrections.

## Statement of need

CA is a workhorse algorithm for exploring contingency tables. However, as the size of the data matrix increases, it often becomes too large to fit into memory. In such cases, an out-of-core (OOC) implementation — where only subsets of data stored on disk are loaded into memory for computation — is desirable. Additionally, representing the data in a sparse matrix format, where only non-zero values and their coordinates are stored, is computationally advantageous. Therefore, a CA implementation that supports both OOC computation and sparse data handling is highly desirable (Figure 1).

Similar discussions have been made in the context of PCA and Non-negative Matrix Factorization (NMF), and we have independently developed Julia packages `OnlinePCA.jl` (Tsuyuzaki et al., 2020; Tsuyuzaki, 2026) and `OnlineNMF.jl` (Tsuyuzaki, 2025). `OnlineCA.jl` is a spin-off version of these, implementing CA.

The standardized residual matrix for CA is

$$S = D_r^{-1/2} (P - r_p c_p^\top) D_c^{-1/2},$$

where  $P = X/n$  is the relative-frequency matrix,  $r_p = P\mathbf{1}$  and  $c_p = P^\top\mathbf{1}$  are the row- and column-mass vectors, and  $D_r = \text{diag}(r_p)$ ,  $D_c = \text{diag}(c_p)$  are the corresponding diagonal mass matrices. `OnlineCA.jl` never explicitly forms  $S$ . Instead, the matrix–vector products  $Sv$  and  $S^\top u$  are computed by streaming through chunks of  $X$  on disk, and the top singular

triplets are extracted with a Halko-style randomized SVD (Halko, Martinsson, & Tropp, 2011; Halko, Martinsson, Shkolnisky, et al., 2011). The same implicit operator works uniformly for dense (CSV), sparse (Matrix Market), and binary (Binary COO) inputs.

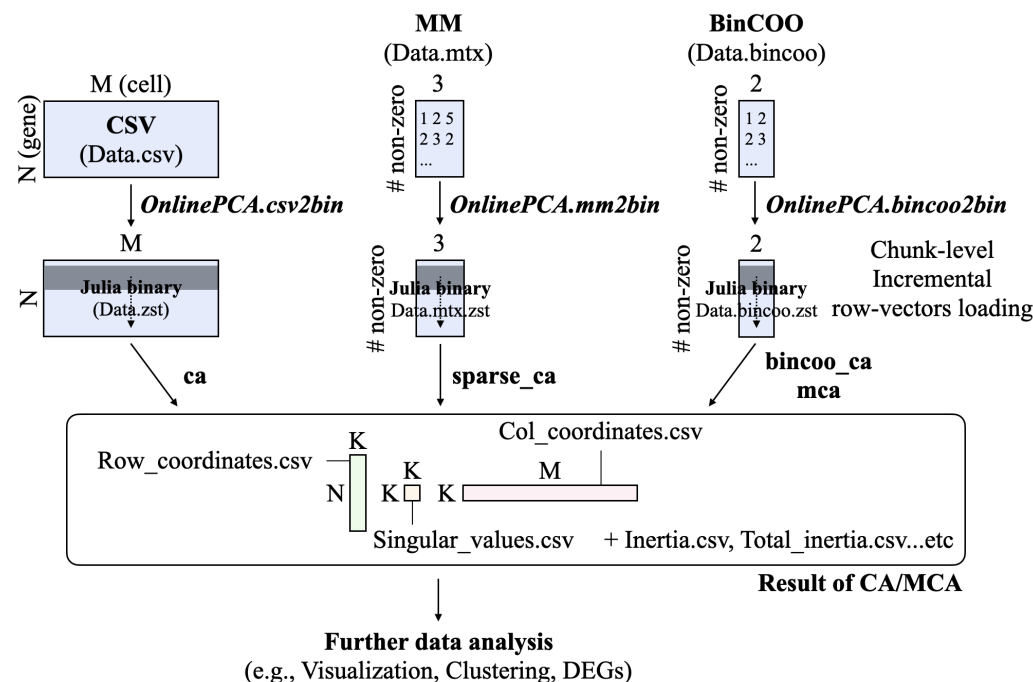


Figure 1: Overview of workflow in OnlineCA.jl.

## Example

CA can be easily reproduced on any machine where Julia is pre-installed by using the following commands in the Julia REPL window:

## Installation

First, install OnlineCA.jl directly from GitHub:

```
# Install OnlinePCA.jl (used for binary I/O preprocessing)
julia> Pkg.add(url="https://github.com/rikenbit/OnlinePCA.jl.git")

# Install OnlineCA.jl
julia> Pkg.add(url="https://github.com/chiba-ai-med/OnlineCA.jl.git")
```

## Preprocess of CSV

Then, write a synthetic data set as a CSV file and convert it to a compressed binary format using Zstandard. Matrix Market (MM) format is also supported for sparse matrices.

```
using OnlinePCA
using OnlinePCA: write_csv
using OnlineCA
using Distributions
using DelimitedFiles
using SparseArrays
using MatrixMarket
```

```
# CSV (input is assumed to be a non-negative integer count matrix)
tmp = mktempdir()
data = Int64.(ceil.(rand(NegativeBinomial(1, 0.5), 300, 99)))
data[1:50, 1:33] .= 100*data[1:50, 1:33]
data[51:100, 34:66] .= 100*data[51:100, 34:66]
data[101:150, 67:99] .= 100*data[101:150, 67:99]
write_csv(joinpath(tmp, "Data.csv"), data)

# Matrix Market (MM)
mmwrite(joinpath(tmp, "Data.mtx"), sparse(data))

# Binarization (Zstandard)
csv2bin(csvfile=joinpath(tmp, "Data.csv"),
        binfile=joinpath(tmp, "Data.zst"))

# Sparsification (Zstandard + MM format)
mm2bin(mmfile=joinpath(tmp, "Data.mtx"),
        binfile=joinpath(tmp, "Data.mtx.zst"))
```

## Plot settings

Define a helper function to visualize the row coordinates of CA using the PlotlyJS.jl package. It generates two subplots: Component-1 vs Component-2 and Component-2 vs Component-3, with color-coded groups.

```
using DataFrames
using PlotlyJS

function subplots(out_ca, group)
    F = out_ca[1]
    data_left = DataFrame(ca1=F[:,1], ca2=F[:,2], group=group)
    data_right = DataFrame(ca2=F[:,2], ca3=F[:,3], group=group)
    p_left = Plot(data_left, x=:ca1, y=:ca2, mode="markers",
                  marker_size=10, group=:group)
    p_right = Plot(data_right, x=:ca2, y=:ca3, mode="markers",
                  marker_size=10, group=:group, showlegend=false)
    for p in (p_left, p_right)
        p.data[1]["marker_color"] = "red"
        p.data[2]["marker_color"] = "blue"
        p.data[3]["marker_color"] = "green"
    end
    p_left.data[1]["name"] = "group1"
    p_left.data[2]["name"] = "group2"
    p_left.data[3]["name"] = "group3"
    p_left.layout["title"] = "Component 1 vs Component 2"
    p_right.layout["title"] = "Component 2 vs Component 3"
    p_left.layout["xaxis_title"] = "ca-1"
    p_left.layout["yaxis_title"] = "ca-2"
    p_right.layout["xaxis_title"] = "ca-2"
    p_right.layout["yaxis_title"] = "ca-3"
    plot([p_left p_right])
end

group = vcat(repeat(["group1"], inner=100),
```

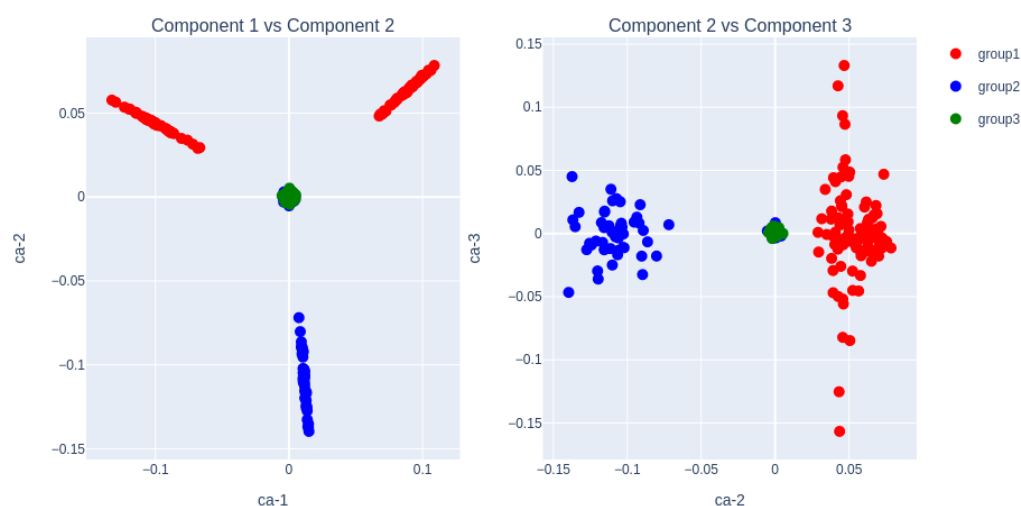
```
repeat(["group2"], inner=100),
repeat(["group3"], inner=100))
```

## CA on dense binary input

This example performs CA on the dense Zstandard-compressed binary file (Figure 2). The standardized residual matrix is never formed explicitly: `ca` streams through the data on disk to compute the implicit operator products and extracts the top `dim` components with randomized SVD.

```
out_ca = ca(input=joinpath(tmp, "Data.zst"),
            dim=3, noversamples=5, niter=3, chunksize=100)
```

```
subplots(out_ca, group)
```



**Figure 2:** Output of `ca` against binarized CSV format.

## Sparse CA on Matrix Market input

This example performs CA on the sparse MM input (Figure 3). `sparse_ca` reads (row, col, value) triplets in chunks and applies the same implicit operator without ever materializing dense intermediates of size  $N \times M$ .

```
out_sparse_ca = sparse_ca(input=joinpath(tmp, "Data.mtx.zst"),
                          dim=3, noversamples=5, niter=3, chunksize=100)
```

```
subplots(out_sparse_ca, group)
```

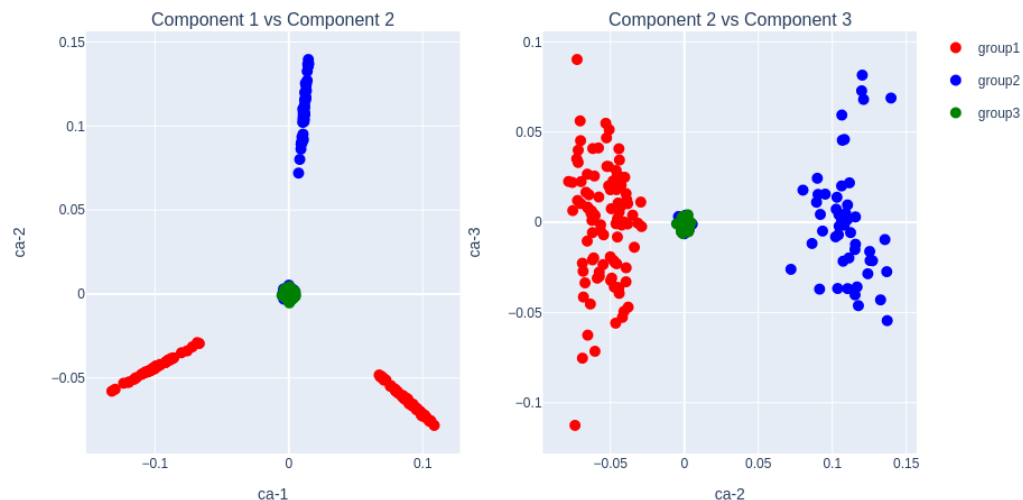


Figure 3: Output of `sparse_ca` against binarized MM format.

## Multiple Correspondence Analysis (MCA)

MCA generalizes CA to a categorical table with  $q$  variables by applying CA to its  $N \times M$  complete disjunctive (indicator) matrix, where each row has exactly  $q$  ones. `OnlineCA.jl` materializes the indicator on disk in BinCOO format and reuses `bincoo_ca`, so all out-of-core, sparse, and randomized-SVD machinery applies unchanged. The CA spectrum on an indicator matrix is artificially inflated; the standard per-axis correction

$$\lambda_k^* = \left( \frac{q}{q-1} \left( \lambda_k - \frac{1}{q} \right) \right)^2 \quad \text{if } \lambda_k > \frac{1}{q}, \text{ else } 0$$

is selectable through the correction keyword (Benzécri (Benzécri, 1979) vs. Greenacre (Greenacre, 2017)).

```
# 200 observations × 4 categorical variables, each with 3 levels
table = rand(1:3, 200, 4)
```

```
out_mca = mca(table; dim=3, correction=:benzecri,
               var_names=["age", "sex", "region", "occupation"])
# out_mca.inertia_adjusted, out_mca.total_inertia_adjusted, ...
```

For more details, including the BinCOO variant for binary contingency tables, see the README.md of `OnlineCA.jl` at <https://github.com/chiba-ai-med/OnlineCA.jl>.

## Related work

There are various implementations of CA / MCA (Esteves, 2017; Halford, 2024; Lê et al., 2008; Nenadić & Greenacre, 2007; Oksanen & others, 2024) and some of them support sparse data formats (DeBruine, 2023; Hsu & Culhane, 2023), but `OnlineCA.jl` is the only tool that supports both OOC computation and language-agnostic sparse formats (MM and BinCOO), enabling seamless integration with external data pipelines.

| Function Name   | Language | OOB | Sparse Format | MCA |
|-----------------|----------|-----|---------------|-----|
| MASS::corresp   | R        | No  | -             | No  |
| ca::ca          | R        | No  | -             | No  |
| ca::mjca        | R        | No  | -             | Yes |
| FactoMineR::CA  | R        | No  | -             | No  |
| FactoMineR::MCA | R        | No  | -             | Yes |
| vegan::cca      | R        | No  | -             | No  |
| prince.CA/MCA   | Python   | No  | -             | Yes |
| mca             | Python   | No  | -             | Yes |
| corral          | R        | No  | dgCMatrix     | No  |
| singlet         | R        | No  | dgCMatrix     | No  |
| OnlineCA.jl     | Julia    | Yes | MM/BinCOO     | Yes |

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